

ZE FAN

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EDUCATION

Johns Hopkins University, Baltimore, MD

2026/09 – now *Ph.D.* in Mathematics
Advisor: Undecided

Wesleyan University, Middletown, CT

2022/09 – 2026/05 *B.A.* in Mathematics with a minor in Economics

PUBLICATIONS

1. Constructive and Stable Cartan–Dieudonné and Applications to Binary Quadratic Forms over Number Fields (joint with Han Li). *Res. Number Theory* **12** (2026), 30. [\[DOI\]](#)

ACADEMIC AWARDS

- May 2026 **Wesleyan University Rice Prize**
Awarded to a senior for excellence in mathematics.
- May 2025 **Wesleyan University Rae Shortt Prize**
Awarded to a junior for excellence in mathematics.
- May 2024 **Wesleyan University Robertson Prize**
Awarded to a sophomore for excellence in mathematics.
- May 2023 **Wesleyan University Sherman Prize-Mathematics**
Awarded to a freshman for excellence in mathematics.
- Dec. 2020 **Award of Excellence, Global Final of S.–T. Yau High School Science Award**
Award winning paper: Preprint [1](#).

TALKS AND SEMINARS

2. *A Constructive and Stable Cartan–Dieudonné Theorem for Binary Quadratic Forms*. Algebra Seminar, Wesleyan University, Sept. 26, 2025.
1. *Lubin–Tate Theorem and Construction of Morava E -Theory*. Participant Talks, University of Chicago math REU, Aug. 7, 2025. [\[Slides\]](#)

OTHER PREPRINTS

4. The Lubin–Tate Deformation Theorem and the Morava E -theory. Expository, Sept. 2025. [\[Preprint\]](#)
3. Stable Regularity Lemmas and their Model-Theoretic Foundations. Expository, Apr. 2025. [\[Preprint\]](#)
2. Sparse Vectors of Small Height Avoiding Hyperplanes. Poster, July 2024. [\[Preprint\]](#)
1. Disjoint Cycles in Ordinary Multipartite Tournaments and Round-Robin Tournaments. High School, Dec. 2020. [\[Preprint\]](#)

¹Updated July 21, 2026